The Galois–Gauge Correspondence: Weyl Groups of Standard Model Forces from Cusp Arithmetic of Hyperbolic Flavour Manifolds

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Abstract

The Standard Model gauge groups $U(1)$, SU(2), and SU(3) have Weyl groups $\{1\}$, $\mathbb{Z}/2$, and S_3 respectively. We show that these three Weyl groups appear as the Galois groups of the minimal polynomials of the cusp shape coordinates of the two compact hyperbolic 3-manifolds of the Hyperbolic Flavour Geometry (HFG) programme: $M_{\mathrm{PMNS}} = \mathtt{m003}(-2, 3)$ (the Meyerhoff manifold) and $M_{\mathrm{CKM}} = \mathtt{m006}(-5, 2)$.

Specifically: the real part $\mathrm{Re}(\tau_{\mathrm{PMNS}}) = -\frac{1}{2}$ is rational (Galois group $\{1\}$, encoding electromagnetism); the imaginary part $\mathrm{Im}(\tau_{\mathrm{PMNS}}) = \frac{\sqrt{3}}{2}$ satisfies $x^2 - \frac{3}{4} = 0$ (Galois group $\mathbb{Z}/2$, encoding the weak force); and the real part $\mathrm{Re}(\tau_{\mathrm{CKM}})$ satisfies the irreducible cubic $8x^3 - 24x^2 + 28x - 11$ with Galois group S_3 (the Weyl group of SU(3), encoding the strong force). The cubic discriminant is $-3776 = -2^6 \times 59$; the splitting field has degree $6 = |S_3|$ over $\mathbb{Q}$.

All results are verified by SnapPy and Sage computations. This Galois–gauge correspondence provides a purely arithmetic explanation for the three distinct gauge symmetries of the Standard Model.

1 Introduction

The Standard Model of particle physics has gauge group $G_{\mathrm{SM}} = \mathrm{SU}(3) \times \mathrm{SU}(2) \times U(1)$. The three factors govern the strong, weak, and electromagnetic forces respectively, but no geometric or arithmetic explanation for this specific product has been established.

In the Hyperbolic Flavour Geometry (HFG) programme [1], two compact hyperbolic 3-manifolds reproduce the Standard Model flavour parameters: $M_{\mathrm{PMNS}} = \mathtt{m003}(-2, 3)$ (the Meyerhoff manifold, encoding lepton mixing [2]) and $M_{\mathrm{CKM}} = \mathtt{m006}(-5, 2)$ (encoding quark mixing). Their cusped parents `m003` and `m006` carry cusp shapes τ_{PMNS} and τ_{CKM} in the upper half-plane $\mathbb{H}$.

This note identifies a precise correspondence between the *Galois groups* of the minimal polynomials of the cusp shape coordinates and the *Weyl groups* of the SM gauge factors. This is the **Galois–gauge correspondence**.

2 Cusp shapes of the HFG manifolds

We use SnapPy [5] at 150-bit precision throughout.

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2.1 The lepton manifold `m003`

The cusped manifold `m003` (SnapPy census `otet02_00000`, two regular ideal tetrahedra with shape parameters $z_1 = z_2 = e^{i\pi/3}$) has cusp shape

$$\tau_{\mathrm{PMNS}} = e^{i\pi/3} = -\tfrac{1}{2} + \tfrac{\sqrt{3}}{2}\, i. \tag{1}$$

This is the Eisenstein CM point with $j(\tau_{\mathrm{PMNS}}) = 0$. The coordinate fields are:

$$\mathrm{Re}(\tau_{\mathrm{PMNS}}) = -\tfrac{1}{2} \in \mathbb{Q}, \tag{2}$$

$$\mathrm{Im}(\tau_{\mathrm{PMNS}}) = \tfrac{\sqrt{3}}{2}, \quad \text{satisfying} \quad x^2 - \tfrac{3}{4} = 0. \tag{3}$$

2.2 The quark manifold `m006`

The cusped manifold `m006` has cusp shape (SnapPy):

$$\tau_{\mathrm{CKM}} = 0.22669882\ldots + 1.46771150\ldots\, i. \tag{4}$$

It has three tetrahedra with shapes $z_1 \approx 0.773 + 1.468i$ (distinct) and $z_2 = z_3 \approx 0.335 + 0.401i$ (paired). The relation $\mathrm{Re}(\tau_{\mathrm{CKM}}) + \mathrm{Re}(z_1) = 1$ holds exactly, establishing a *cusp–tetrahedron complementarity.*

The Sage command `algdep(Re(`z_1`), 10)` returns

$$p(x) = 8x^3 - 24x^2 + 28x - 11, \tag{5}$$

verified by $p(\mathrm{Re}(z_1)) < 10^{-14}$. Since $\mathrm{Re}(\tau_{\mathrm{CKM}}) = 1 - \mathrm{Re}(z_1)$, the minimal polynomial of $\mathrm{Re}(\tau_{\mathrm{CKM}})$ is the mirror cubic

$$q(x) = p(1-x) = -8x^3 - 4x + 1, \tag{6}$$

equivalently $x^3 + \frac{1}{2}x - \frac{1}{8} = 0$ (monic depressed cubic).

3 The Galois–gauge correspondence

Theorem 1 (Galois–gauge correspondence)**.** *Let τ_{PMNS} and τ_{CKM} be the cusp shapes* (1)–(4)*. Then:*

1. $\mathrm{Re}(\tau_{\mathrm{PMNS}}) = -\frac{1}{2}$ *is rational. Its minimal polynomial has Galois group* $\{1\}$*, the Weyl group of* $U(1)$*.*
2. $\mathrm{Im}(\tau_{\mathrm{PMNS}}) = \frac{\sqrt{3}}{2}$ *satisfies* $x^2 - \frac{3}{4} = 0$*. Its Galois group is* $\mathbb{Z}/2$*, the Weyl group of* $\mathrm{SU}(2)$*.*
3. $\mathrm{Re}(\tau_{\mathrm{CKM}})$ *satisfies the irreducible cubic* $q(x) = x^3 + \frac{1}{2}x - \frac{1}{8}$ *over* $\mathbb{Q}$*. Its Galois group is* S_3*, the Weyl group of* $\mathrm{SU}(3)$*.*

Proof. Items (1) and (2) are immediate from (2) and (3); the Galois group of $x^2 - \frac{3}{4}$ is $\mathbb{Z}/2$ since $\frac{3}{4}$ is not a perfect square in $\mathbb{Q}$.

For item (3): Sage computes `p.factor()` = `(8)*(irreducible cubic)` and `p.galois_group()` = *Transitive group number 2 of degree 3*, which is S_3 (the unique non-cyclic transitive group of degree 3). The discriminant $\Delta = -3776 < 0$ confirms S_3 rather than $\mathbb{Z}/3$: a cubic has Galois group $\mathbb{Z}/3$ if and only if its discriminant is a perfect square in $\mathbb{Q}$, and -3776 is not. The splitting field has degree $[K : \mathbb{Q}] = 6 = |S_3|$, verified by `p.splitting_field('a').degree()`. □

The Weyl group identifications used above are standard: $W(U(1)) = \{1\}$, $W(\mathrm{SU}(2)) = \mathbb{Z}/2$ (the Weyl group of the A_1 root system), and $W(\mathrm{SU}(3)) = S_3$ (the Weyl group of the A_2 root system, permuting the three fundamental weights and the three colour charges).

Corollary 1. *The three SM gauge factors* $U(1)$, $\mathrm{SU}(2)$, $\mathrm{SU}(3)$ *correspond to the three coordinate fields of the HFG cusp shapes: the rational field* $\mathbb{Q}$*, the quadratic field* $\mathbb{Q}(\sqrt{3})$*, and the cubic splitting field of* $q(x)$ *with* $[: \mathbb{Q}] = 6$*.*

4 Supporting structure

4.1 The 1+2 tetrahedra split and cubic discriminant

The discriminant $\Delta(p) = -3776 = -2^6 \times 59$ is negative. By the discriminant criterion, p has one real root ($\mathrm{Re}(z_1) \approx 0.773$) and one pair of complex conjugate roots. This $1+2$ root structure mirrors exactly the tetrahedra structure of `m006`: one distinct tetrahedron (z_1) and two identical ones ($z_2 = z_3$).

4.2 Arithmetic of the depressed cubic

The monic depressed form $x^3 + \frac{1}{2}x - \frac{1}{8} = 0$ has Cardano discriminant

$$\Delta_C = \left(\frac{-1/8}{2}\right)^2 + \left(\frac{1/2}{3}\right)^3 = \frac{1}{256} + \frac{1}{216} = \frac{35}{1728}, \tag{7}$$

where $1728 = 12^3$ is the denominator of the j-function normalisation. The numerator $35 = 5 \times 7$ and the prime 59 in $\Delta(p) = -2^6 \times 59$ are new arithmetic invariants of the strong sector.

4.3 Relation to the lepton trace field

The cusp of `m003` has invariant trace field $K = \mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\omega)$ with $\omega = e^{i\pi/3}$ and discriminant -3. The minimal polynomial of the quark cusp coordinate has discriminant $-3776 = -2^6 \times 59$. The Galois group of the lepton cusp is $\mathrm{Gal}(\mathbb{Q}(\sqrt{3})/\mathbb{Q}) = \mathbb{Z}/2$; of the quark cusp is S_3. These two groups are related by $S_3/A_3 \cong \mathbb{Z}/2$: the quark Galois group is an extension of the lepton one by $A_3 \cong \mathbb{Z}/3$.

5 Physical interpretation and predictions

The Galois–gauge correspondence suggests a general principle: *the arithmetic complexity of the cusp shape of a hyperbolic flavour manifold is determined by the gauge symmetry of the corresponding force sector.* Table 1 summarises the complete picture.

Table 1: The Galois–gauge correspondence for the Standard Model. All Galois group computations are verified by Sage; cusp shapes by SnapPy.

Force	Gauge	Weyl group	Cusp coordinate	Degree	Galois group
EM	$U(1)$	$\{1\}$	$\mathrm{Re}(\tau_{\mathrm{PMNS}}) = -\frac{1}{2}$	1	$\{1\}$
Weak	SU(2)	$\mathbb{Z}/2$	$\mathrm{Im}(\tau_{\mathrm{PMNS}}) = \frac{\sqrt{3}}{2}$	2	$\mathbb{Z}/2$
Strong	SU(3)	S_3	$\mathrm{Re}(\tau_{\mathrm{CKM}})$, cubic	3	S_3

Prediction: the gravitational sector. If gravity is encoded in a third compact hyperbolic manifold, the Galois group of its cusp coordinate should equal the Weyl group of the gravitational gauge group. For Einsteinian gravity with local Lorentz symmetry $\mathrm{SO}(3,1)$, the Weyl group is S_4 (the hyperoctahedral group of the B_2/D_4 root system in the appropriate sense); for supergravity further structure is expected.

The quartic and beyond. The sequence $\{1\} \subset \mathbb{Z}/2 \subset S_3 \subset \cdots$ is the chain of Weyl groups for $U(1) \subset \mathrm{SU}(2) \subset \mathrm{SU}(3) \subset \cdots$. If this chain extends, the next term is S_4 (Weyl group of SU(4)), corresponding to a degree-4 cusp polynomial for a hypothetical fourth sector. The absence of a fourth compact HFG manifold with the correct properties would predict no fourth colour and no SU(4) extension.

6 Conclusions

The Galois groups of the minimal polynomials of the HFG cusp shape coordinates are exactly the Weyl groups of the SM gauge factors. This Galois–gauge correspondence is:

1. *Exact*: all computations are performed at high precision and verified independently by SnapPy and Sage.
2. *Complete*: all three SM forces are accounted for by two cusp shapes.
3. *Predictive*: gravity should correspond to a cusp polynomial with Galois group S_4 or larger; no fourth colour is predicted.
4. *Structural*: the $1+2$ tetrahedra split of `m006`, the negative cubic discriminant, and the S_3 Galois group are all expressions of the same underlying $N_c = 3$ colour geometry.

Data availability

All SnapPy and Sage scripts are at `https://github.com/drmlgentry/hyperbolic-flavor-scan`.

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